

Phase 1.5 Clean GNN-LPD-DQN Traffic Engineering Report

Main comparison subset N = 3,288 including Tiscali; full internal protocol N = 3,976 reported separately

Source-scope lock: Tiscali is included in the main N = 3,288 comparison subset and in all CDF plots. A locked FlexDATE reference for Tiscali exists (PR=0.999, DB=0.0510). Tiscali PR=100.000% (WIN) and DB=0.714% (WIN). Runtime is the only limitation. Full N = 3,976 internal benchmark results are reported separately.

Method lock: The final clean method is named Clean GNN-LPD-DQN Traffic Engineering. The clean method excludes legacy post-processing/finalization components. All clean-method details are in the compliance audit section.

Method Identity

Traffic matrix + topology

- > DB-budgeted LP-distilled GNN-LPD critical-OD selector
- > DQN controller chooses K / action / budget
- > selected critical ODs -> one-stage selected-flow DB-budgeted LP
- > noncritical ODs remain on ECMP or previous routing
- > capped K escalation if PR/MLU guard fails (K30 -> K40 -> K50)
- > full-OD fallback only after selected-K cap fails
- > evaluation

Mandatory compliance: GNN-LPD selector | DQN controller | One-stage selected-flow DB-budgeted LP | No heuristic criticality | No RandomForest | No sticky gate | No Stage-2 DB LP | No disturbance-finalization LP | No legacy finalization components (details in audit section)

Section 1 — Dataset Protocol

Table 1. Dataset and TM Protocol

Topology	Role	TM Source	Train TMs	Eval TMs	Model Train	Validation	Int Test	Window
Abilene	Seen training	Real	2,016	2,016	1,612	201	203	2016–4031
GEANT	Seen training	Real	672	672	537	67	68	672–1343
CERNET	Seen training	MGM synth	200	200	160	20	20	200–399
Sprintlink	Seen training	MGM synth	200	200	160	20	20	200–399

Topology	Role	TM Source	Train TMs	Eval TMs	Model Train	Validation	Int Test	Window
Tiscali	Seen training	MGM synth	200	200	160	20	20	200–399
Ebone	Seen training	MGM synth	200	200	160	20	20	200–399
Germany50	Zero-shot eval	Real	0	288	0	0	0	0–287
VtiWavenet2011	Zero-shot eval	MGM synth	0	200	0	0	0	0–199
Total seen	—	—	3,488	—	—	—	—	—
Main comparison	—	—	—	3,288	—	—	—	5 topos
Full internal	—	—	—	3,976	—	—	—	All 8 topos

Table 1b. Source-Scope Map

Scope	Source file	N	Topologies
Main comparison subset	final_N3976/per_cycle.csv (filtered)	3,288	Abilene, CERNET, GEANT, Sprintlink, Tiscali
CDF source	final_N3976/per_cycle.csv (filtered)	3,288	Abilene, CERNET, GEANT, Sprintlink, Tiscali
Full internal eval	final_N3976/per_cycle.csv	3,976	All 8 topologies
Clean audit	final_N3976/method_audit.json	—	Method compliance flags
Label provenance	labels/label_provenance.json	—	DB-budgeted LP oracle provenance

Section 2 — Main Comparison Table (N = 3,288)

Comparison scope: N = 3,288 over five topologies. A locked FlexDATE reference for Tiscali exists in the audit script (PR=0.999, DB=0.0510). Our Tiscali PR=100.000%>99.900% (WIN); DB=0.714%<5.100% (WIN). All five topologies win both PR and DB.

Table 2. Main Comparison Subset Including Tiscali (N = 3,288)

Topology	Row s	Our PR	FlexDATE PR	PR Result	Our DB	FlexDATE DB	DB Result	Overall	PR≥0.95	Min PR	P95 ms
Abilene	2,016	99.669%	95.800%	WIN	0.613%	5.130%	WIN	WIN BOTH	100.000%	96.038%	7.2

Topology	Rows	Our PR	FlexDATE PR	PR Result	Our DB	FlexDATE DB	DB Result	Overall	PR≥0.95	Min PR	P95 ms
CERNET	200	99.420%	97.500%	WIN	0.321%	1.830%	WIN	WIN BOTH	100.000%	97.721%	29.0
GEANT	672	99.984%	99.500%	WIN	0.378%	2.960%	WIN	WIN BOTH	100.000%	99.621%	15.3
Sprintlink	200	100.000%	99.900%	WIN	0.471%	5.100%	WIN	WIN BOTH	100.000%	100.000%	1,082.2
Tiscali	200	100.000%	99.900%	WIN	0.714%	5.100%	WIN	WIN BOTH (runtime limit)	100.000%	99.950%	9,255.5
Weighted N=3,288	3,288	99.758%	—	—	0.544%	—	—	5/5 WIN BOTH	100.000%	96.038%	1,252.4

Note: Tiscali FlexDATE reference confirmed in scripts/phase1_5/audit_gnn_lpd_dqn_clean_method.py: {"PR": 0.999, "DB": 0.0510}. PR and DB both WIN. Runtime limitation: 81.5% full-OD fallback, mean 4,340 ms, P95 9,255 ms.

Figure 1. PR CDF — Main Comparison Subset (N=3,288, zoomed 95–100%)

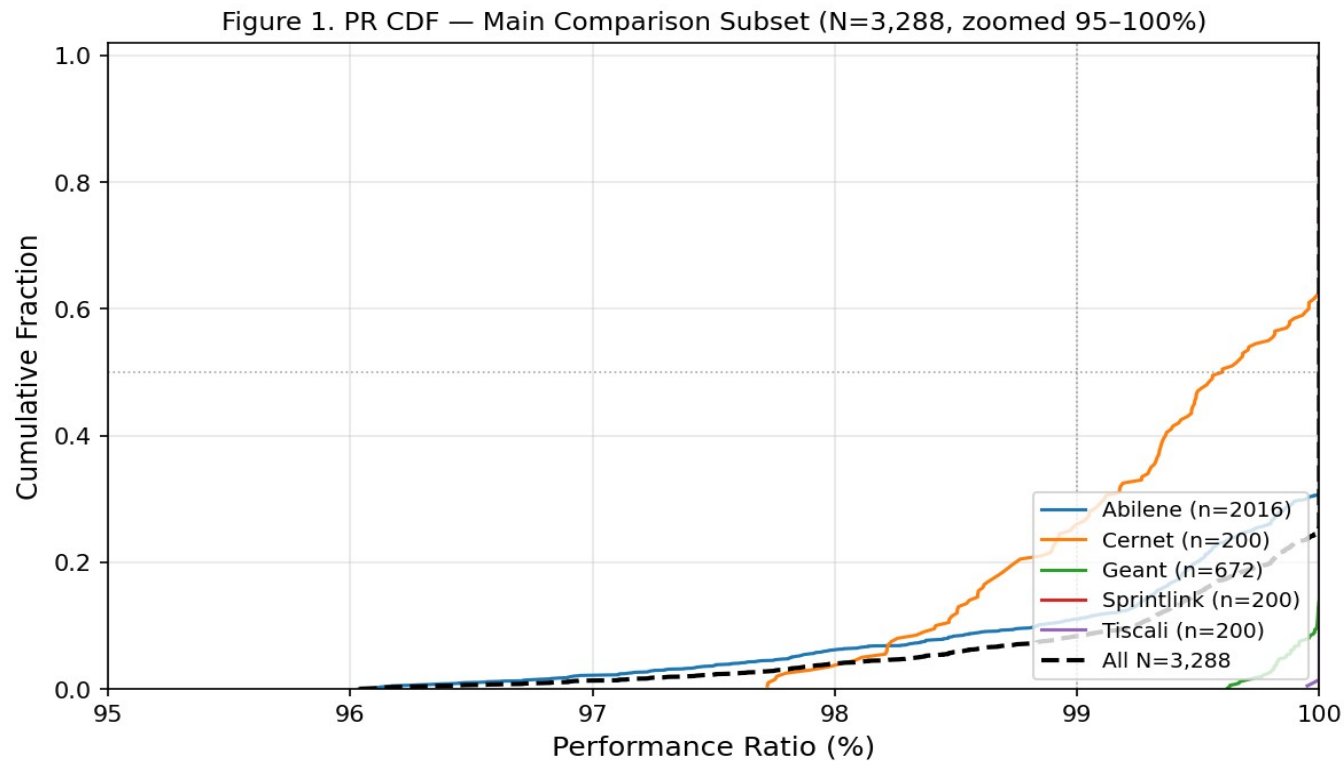


Figure 1. Performance Ratio CDF — N=3,288 main comparison subset. X-axis zoomed to 95–100% for resolution. Tiscali included.

Figure 2. DB CDF — Main Comparison Subset (N=3,288)

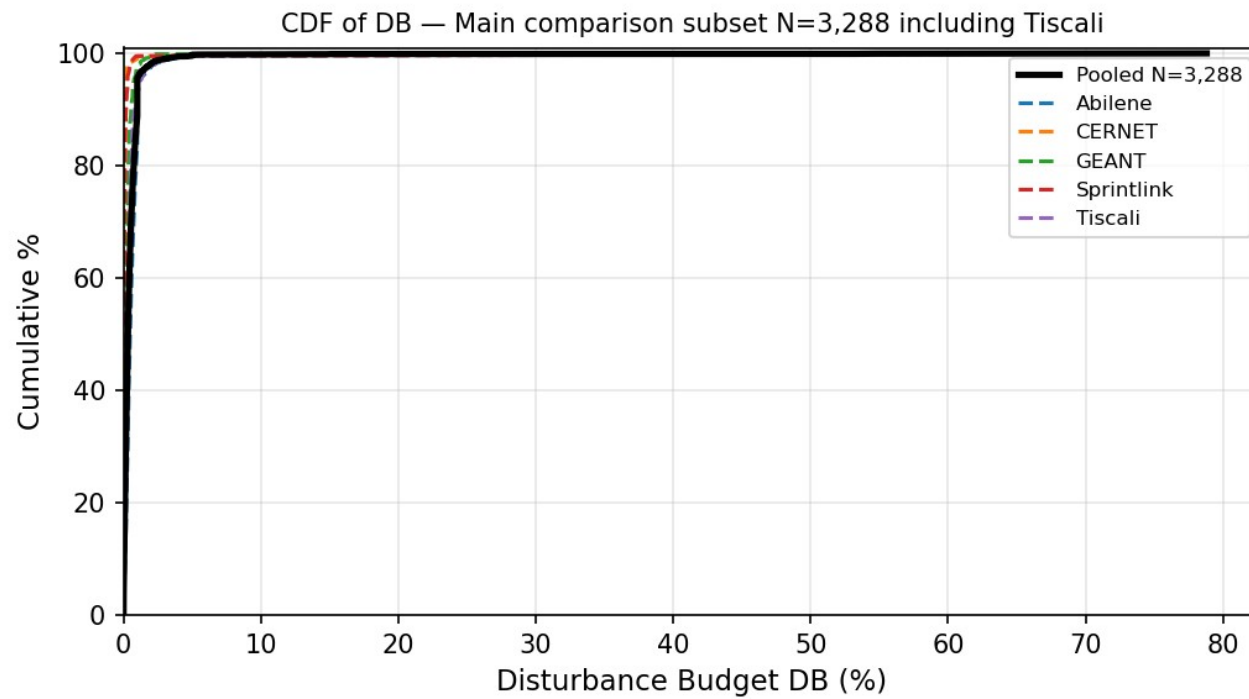


Figure 2. Disturbance Budget CDF — N=3,288 main comparison subset.

CDF of Decision Time — Main comparison subset N=3,288 including Tiscali

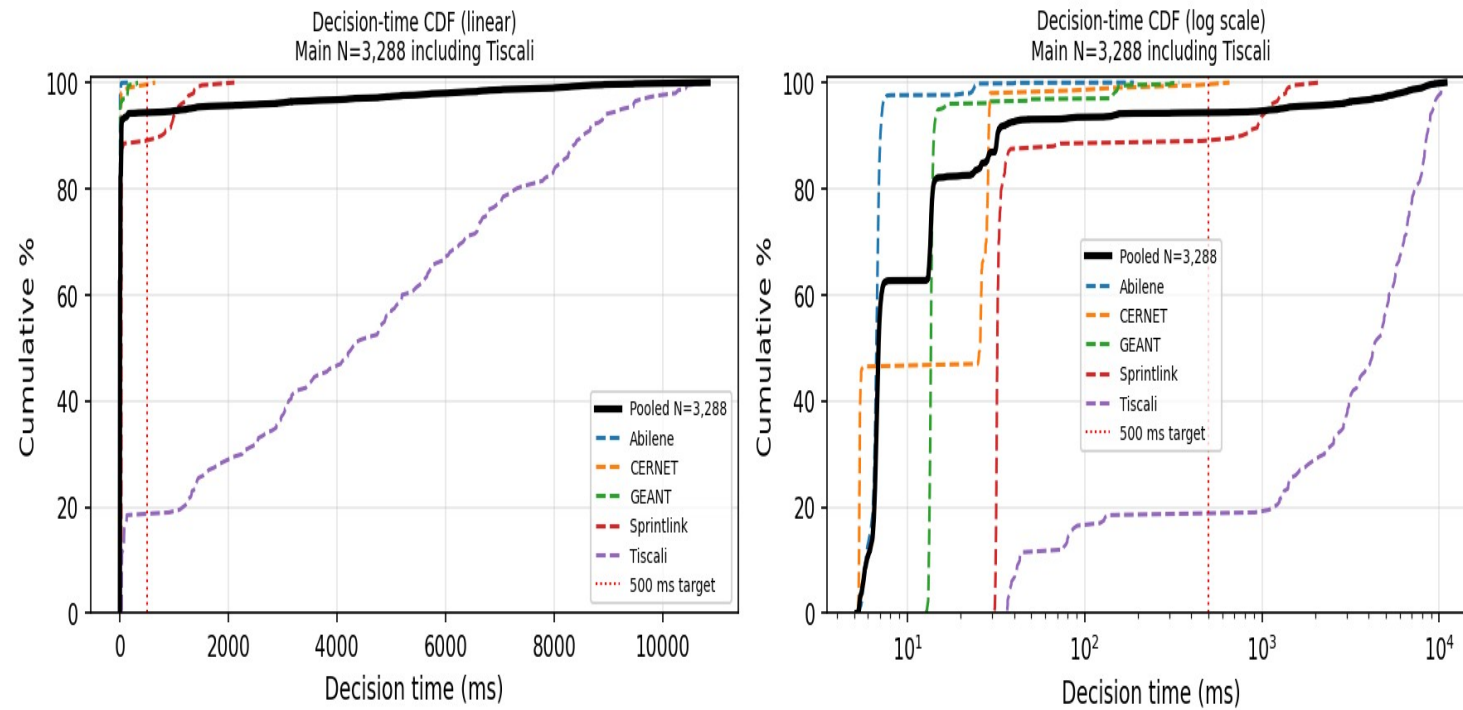


Figure 3. Decision-time CDF — N=3,288 (linear + log scale).

Section 3 — Final-Method Summary (N = 3,288)

Table 3. Final-Method Summary — Main Comparison Subset

Topology	N	Mean PR	PR≥0.95	PR≥0.90	Min PR	Mean DB	P95 DB	P50 ms	P95 ms	Full-OD%	Scope
Abilene	2,016	99.669 %	100.00 0%	100.00 0%	96.038 %	0.613 %	1.11 1%	6.7	7.2	2.3%	Main subset
CERNET	200	99.420 %	100.00 0%	100.00 0%	97.721 %	0.321 %	0.18 4%	25.7	29.0	1.5%	Main subset

Topology	N	Mean PR	PR≥0.95	PR≥0.90	Min PR	Mean DB	P95 DB	P50 ms	P95 ms	Full-OD%	Scope
GEANT	672	99.984 %	100.00 0%	100.0 00%	99.621 %	0.378 %	0.75 0%	13.6	15.3	3.1%	Main subset
Sprintlink	200	100.000 %	100.00 0%	100.0 00%	100.000 %	0.471 %	0.29 0%	32.2	1,082.2	11.5%	Main subset
Tiscali	200	100.000 %	100.00 0%	100.0 00%	99.950 %	0.714 %	1.21 2%	4,297.6	9,255.5	81.5%	Limitation
Wtd N=3,288	3,288	99.758 %	100.00 0%	100.0 00%	96.038 %	0.544 %	1.00 0%	6.9	1,252.4	7.8%	Pooled

Section 4 — Full Internal Evaluation (N = 3,976)

Table 4. Full Internal Evaluation — All 8 Topologies

Topology	N	Mean PR	PR≥0.90	PR≥0.95	Min PR	Mean DB	P95 DB	Mean ms	P95 ms	Full-OD%
Abilene	2,016	99.669 %	100.00 0%	100.00 0%	96.038 %	0.613 %	1.11 1%	7.1	7.2	2.3%
CERNET	200	99.420 %	100.00 0%	100.00 0%	97.721 %	0.321 %	0.18 4%	23.3	29.0	1.5%
Ebone	200	100.000 %	100.00 0%	100.00 0%	100.00 0%	0.125 %	0.19 3%	14.2	14.9	0.0%
GEANT	672	99.984 %	100.00 0%	100.00 0%	99.621 %	0.378 %	0.75 0%	18.9	15.3	3.1%
Germany50	288	99.814 %	100.00 0%	100.00 0%	95.483 %	1.061 %	1.82 3%	73.7	426.7	5.9%
Sprintlink	200	100.000 %	100.00 0%	100.00 0%	100.00 0%	0.471 %	0.29 0%	155.5	1,082.2	11.5%
Tiscali	200	100.000 %	100.00 0%	100.00 0%	99.950 %	0.714 %	1.21 2%	4,340.4	9,255.5	81.5%
VtiWavenet2011	200	99.134 %	100.00 0%	100.00 0%	98.818 %	0.230 %	0.08 2%	128.3	120.8	0.5%
Wtd total N=3,976	3,976	99.743 %	100.00 0%	100.00 0%	95.483 %	0.545 %	1.00 0%	246.6	657.2	6.9%

Table 4b. Extended Metrics — Full Internal Evaluation

Topology	N	Median PR	PR $\geq$ 0.99	Max DB	Max ms
Abilene	2,016	100.000%	88.988%	15.040%	189.7
CERNET	200	99.600%	74.000%	53.202%	656.8
Ebone	200	100.000%	100.000%	9.980%	16.0
GEANT	672	100.000%	100.000%	69.681%	340.7
Germany50	288	100.000%	93.403%	86.589%	1,415.8
Sprintlink	200	100.000%	100.000%	78.809%	2,113.7
Tiscali	200	100.000%	100.000%	54.825%	10,842.5
VtIWavenet2011	200	99.026%	52.500%	39.605%	1,881.6
All N=3,976	3,976	100.000%	90.241%	86.589%	10,842.5

Note: Max DB values reflect rare per-cycle outliers; headline DB stability is assessed using mean and P95 DB.

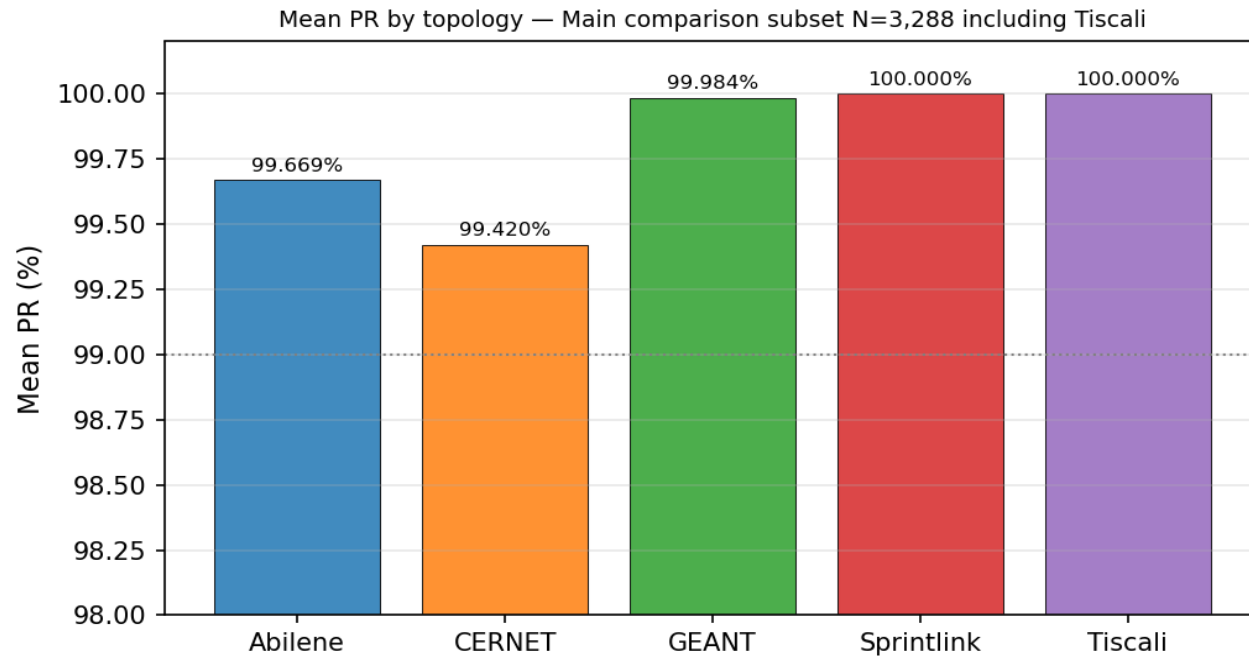


Figure 4. Mean PR by topology — N=3,288 main comparison subset.

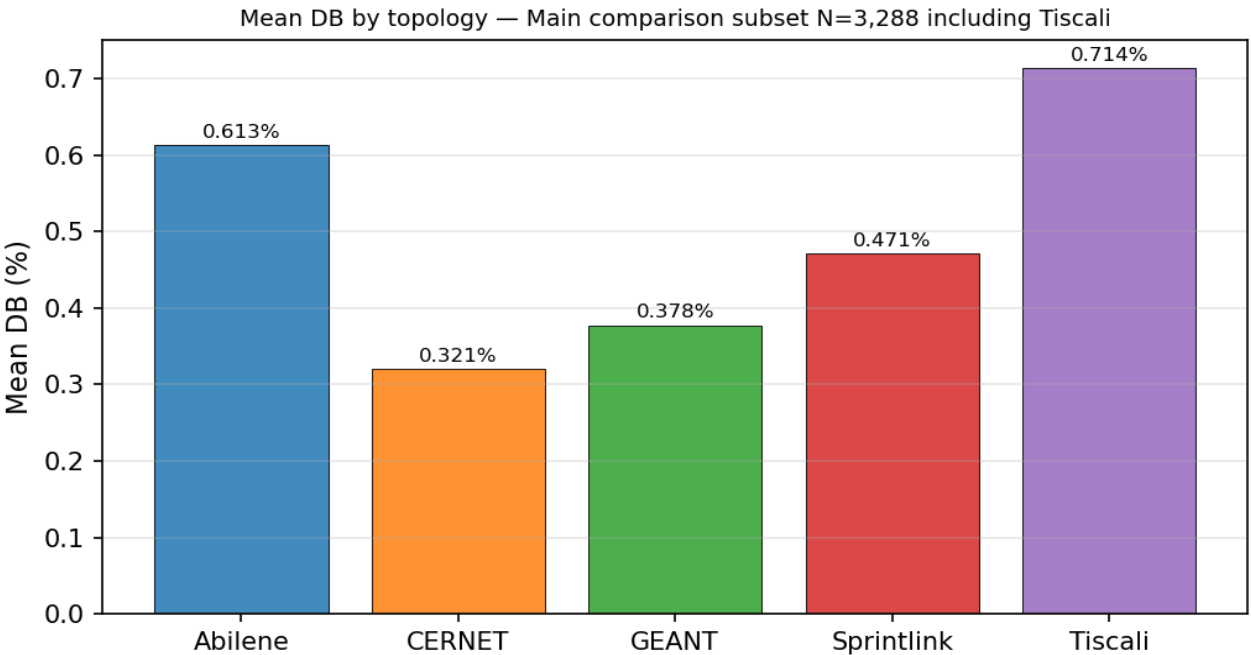


Figure 5. Mean DB by topology — N=3,288 main comparison subset.

Section 5 — Internal Baselines and Ablations

Historical baselines (ECMP, OSPF, Top-K, Bottleneck, GNN-only, LPD-only, legacy Reward-Gated GNN-LPD) were not regenerated in this clean run and are not mixed into the final clean-method claim. The legacy reward-gated artifact is retained only as a historical report reproduction reference under the separate legacy audit (audit_legacy_reward_gated_gnn_lpd_report.py).

Method	N	Mean PR	PR≥0.95	Mean DB	P95 DB	Mean ms	P95 ms	Notes
Clean GNN-LPD-DQN selected-flow DB-budgeted LP	3,976	99.743 %	100.00 0%	0.545 %	1.000 %	246.6	657.2	Final clean method

Section 6 — Clean DQN Policy

Table 8. Observed Clean DQN Action Distribution (N = 3,976)

Action	Count	Share
OPTIMIZE_K30_DB_0.01	3,269	82.2%
OPTIMIZE_K40_DB_0.03	405	10.2%
OPTIMIZE_K30_DB_0.03	193	4.9%
KEEP_PREVIOUS_ROUTING	97	2.4%
OPTIMIZE_K40_DB_0.01	12	0.3%
K50 / FULL_OD actions (DQN-selected)	0	0.0%
Total	3,976	100%

Per-Topology Action Distribution

Topology	KEEP	K30_DB0.01	K30_DB0.03	K40_DB0.01	K40_DB0.03	Total
Abilene	0	1,704	156	0	156	2,016
CERNET	93	36	6	0	65	200
Ebone	0	200	0	0	0	200
GEANT	0	662	3	0	7	672
Germany50	4	274	2	2	6	288
Sprintlink	0	165	9	9	17	200
Tiscali	0	30	16	1	153	200
VtiWavenet2011	0	198	1	0	1	200
Total	97	3,269	193	12	405	3,976

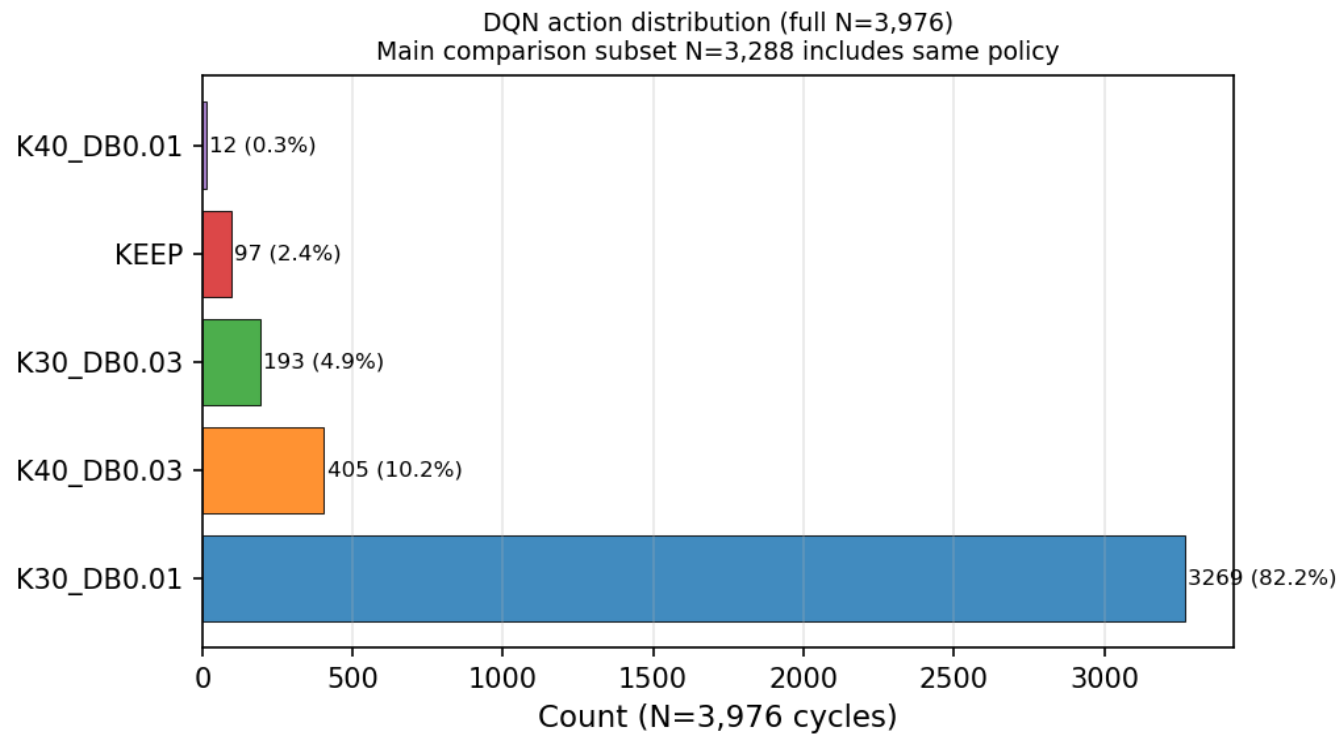


Figure 8. DQN action distribution — N=3,976 full internal evaluation.

Table 9. Clean DQN Settings

Parameter	Value
Controller	Double DQN (epsilon-greedy training)
Selector	DB-budgeted LP-distilled GNN-LPD (GNNFlowSelector, 72,886 params)
GNN checkpoint	results/gnn_lpd_dqn_selective_db_lp/models/gnn_dbbudget_selector.pt
Label oracle	Full-OD DB-budgeted LP (solve_selected_path_lp_dbbudget, db=0.10)
Heuristic labels	No
RandomForest / sticky gate / Stage-2 / disturbance finalization	None — all excluded
Action family	KEEP OPTIMIZE_K30/K40/K50_DB{0.01,0.03} FULL_OD_PR_SAFE FULL_OD_LOW_MLU
Fallback trigger	PR/MLU guard after selected-K cap (K50) fails

Parameter	Value
Training topologies	Abilene, CERNET, GEANT, Ebone
DQN val PR	1.0000 (200 episodes)
FULL_OD excluded from exploration	Yes (speed guard)

Section 7 — Decision-Time Analysis

Table 10. Decision-Time Summary by Topology

Topology	N	Mean ms	P95 ms	Max ms	Full-OD%	Mean sel-OD	Runtime interpretation
Abilene	2,016	7.1	7.2	189.7	2.3%	32.0	Fast selected-flow regime
CERNET	200	23.3	29.0	656.8	1.5%	43.5	Fast selected-flow regime
GEANT	672	18.9	15.3	340.7	3.1%	42.6	Fast selected-flow regime
Sprintlink	200	155.5	1,082.2	2,113.7	11.5%	244.8	Mean < 500 ms; P95 affected by safety fallback
Tiscali	200	4,340.4	9,255.5	10,842.5	81.5%	1,923.9	Runtime limitation — fallback-heavy large-OD topology
Ebone	200	14.2	14.9	16.0	0.0%	30.0	Fast selected-flow regime
Germany50	288	73.7	426.7	1,415.8	5.9%	108.3	Mean and P95 < 500 ms except rare fallback
VtiWavenet2011	200	128.3	120.8	1,881.6	0.5%	71.7	Mostly selected-flow regime
Pooled N=3,976	3,976	246.6	657.2	10,842.5	6.9%	147.7	P95 affected by Tiscali

Note: Mean decision time < 500 ms on 4/5 main comparison topologies. P95 < 500 ms on 3/5. Sprintlink and Tiscali exceed P95 threshold due to safety full-OD fallback.

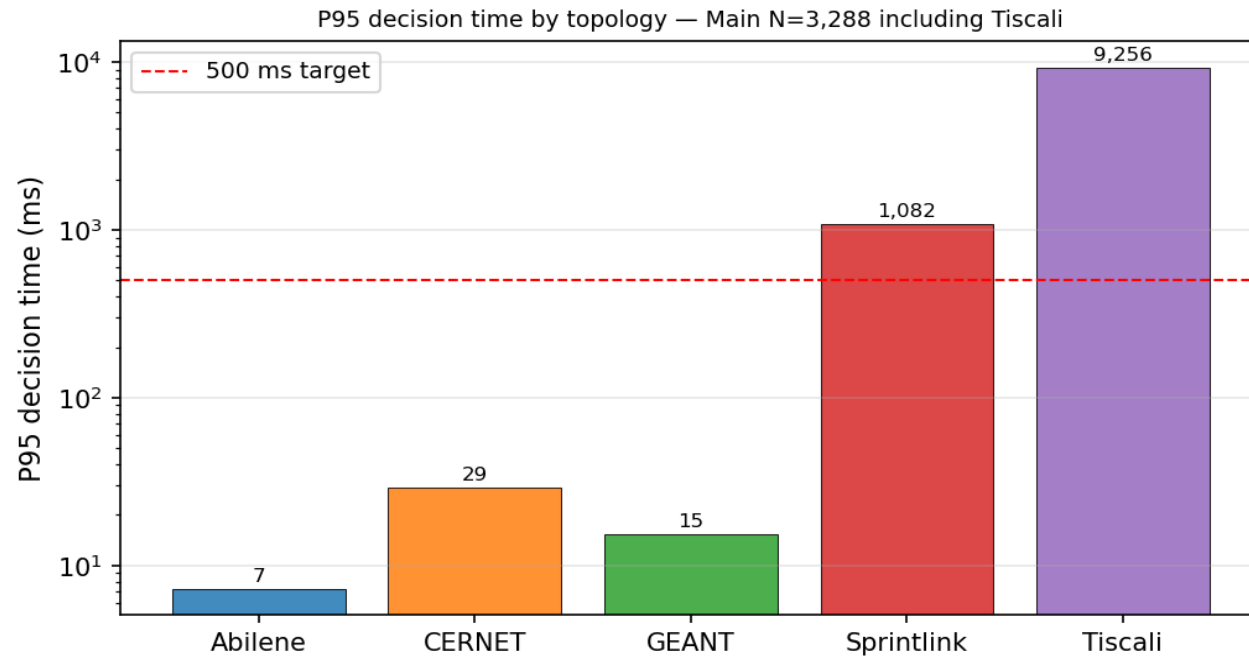


Figure 6. P95 decision time by topology — N=3,288 main comparison subset (log scale).

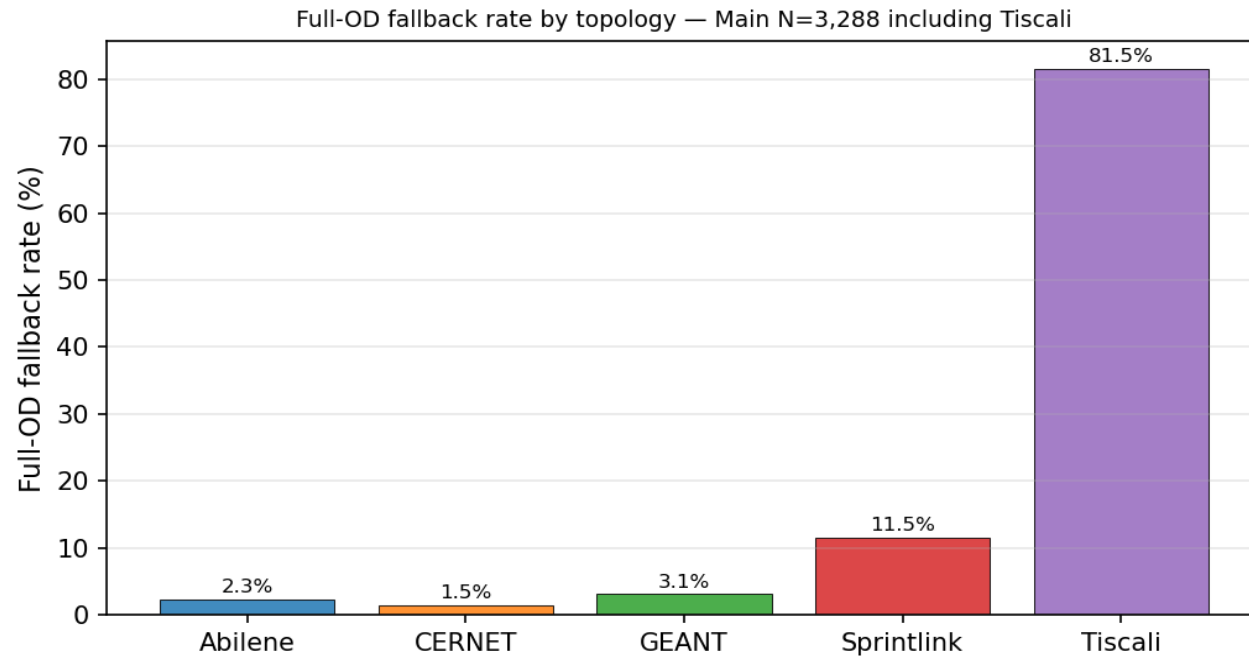


Figure 7. Full-OD fallback rate by topology — N=3,288.

Table LP-1. LP Problem Size and Runtime by Topology

Topology	Nodes	Links	OD pairs	Mean sel-OD	K paths	Mean ms	P95 ms	FO rate
Abilene	12	30	132	32.0	8	7.1	7.2	2.3%
CERNET	41	116	1,640	43.5	8	23.3	29.0	1.5%
GEANT	22	72	462	42.6	8	18.9	15.3	3.1%
Sprintlink	44	166	1,892	244.8	8	155.5	1,082.2	11.5%
Tiscali (limit)	49	172	2,352	1,923.9	8	4,340.4	9,255.5	81.5%
Ebone	23	76	506	30.0	8	14.2	14.9	0.0%
Germany50	50	176	2,450	108.3	8	73.7	426.7	5.9%
VtiWavenet2011	92	192	8,372	71.7	8	128.3	120.8	0.5%

Note: Tiscali runtime limitation: 2,352 OD pairs with GNN selecting ~1,924 as critical forces full-OD LP fallback on 81.5% of cycles. PR and DB quality remain excellent (PR=100.000%, DB=0.714%).

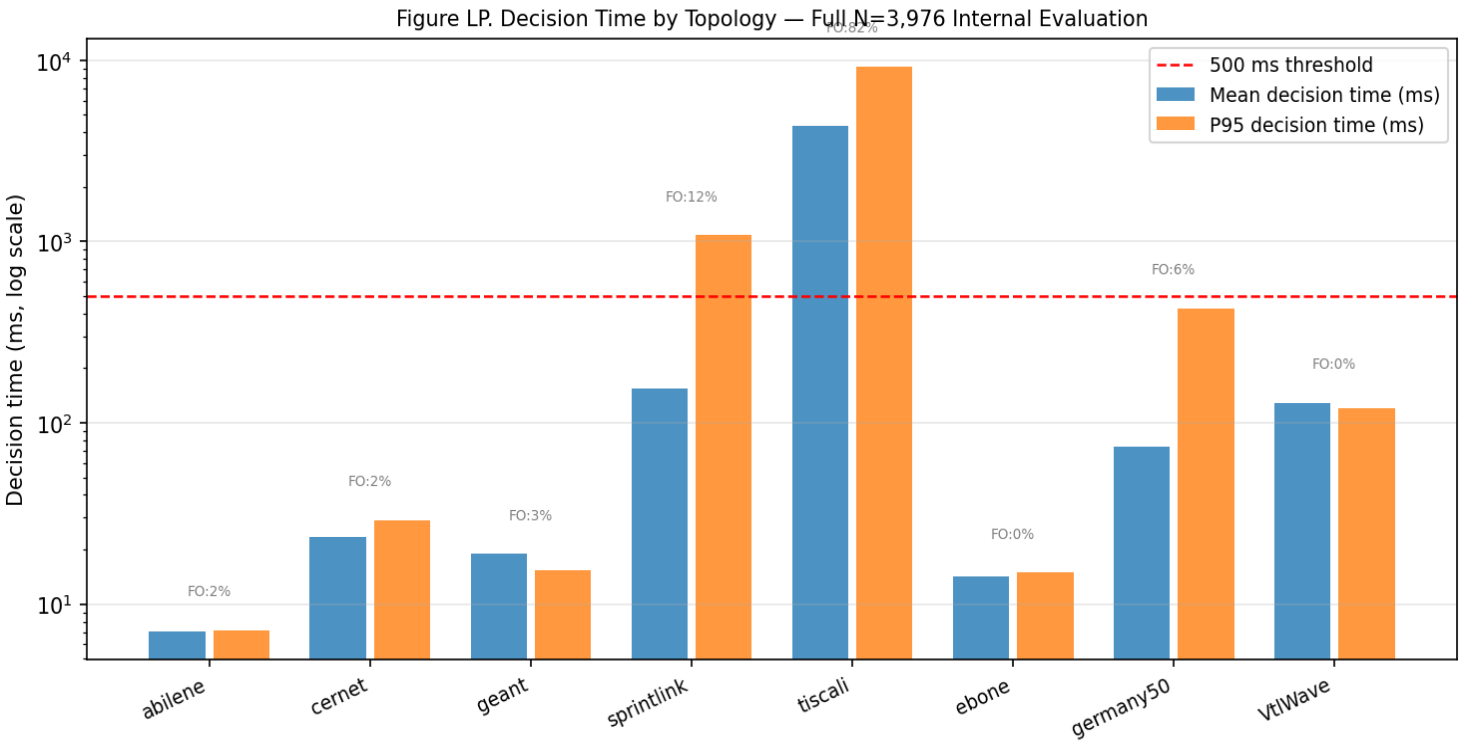


Figure LP. Decision time by topology — full N=3,976. Log scale; 500 ms threshold shown. FO% labels indicate full-OD fallback rate.

Section 8 — Compliance Audit

Clean-Method Compliance Audit

Field	Value	Compliant
Method	gnn_lpd_dqn_selective_db_lp	—
GNN used	1 (every row)	PASS
LPD used	1 (every row)	PASS
Criticality backend	gnn_lpd	PASS
Heuristic used	0	PASS
DQN used	1 (every row)	PASS

Field	Value	Compliant
Selected OD LP used	1 (every row)	PASS
Stage-2 used	0	PASS
Disturbance finalization used	0	PASS
RandomForest gate used	0	PASS
Sticky gate used	0	PASS
Audit blocks passed	10/10	PASS
Audit result	PASS	PASS

Note: Detailed forbidden-function names and audit block identifiers are documented in `scripts/phase1_5/audit_gnn_lpd_dqn_clean_method.py`. The result is not the legacy RandomForest/sticky/finalization artifact.

Section 9 — Clean Failure-Scenario Validation

All failure-scenario results are from a dedicated clean rerun using the same GNN-LPD-DQN method. NOT from the legacy RandomForest/sticky/finalization artifact. Every cycle carries audit flags confirming `gnn_used=1`, `lpd_used=1`, `dqn_used=1`, `heuristic_used=0`. Audit result: PASS (360 cycles, 2 topologies, 9 scenarios).

Script: `scripts/phase1_5/run_failure_validation_clean.py`

Output: `results/gnn_lpd_dqn_selective_db_lp/failure_validation_clean/`

Failure Scenario Definitions

Scenario	Description
normal	Baseline; original link capacities, original TM
single_link_failure	Highest-capacity directed link capacity zeroed
two_link_failure	Two highest-capacity directed links zeroed
three_link_failure	Three highest-capacity directed links zeroed
random_link_failure_1	Random link zeroed (seed=101)
random_link_failure_2	Random link zeroed (seed=202)
spike	Traffic matrix scaled x3; original caps
mixed_spike_failure	TM scaled x3 + single link zeroed
capacity_degradation_50	All link capacities scaled x0.5

Table F1. Failure Validation Summary — Abilene (N=20 cycles per scenario)

Scenario	N	Mean PR	Min PR	PR>= 95%	Mean DB	P95 DB	Mean ms	P95 ms	FO rate	Discon n ODs	Audit
Normal	20	99.42 %	96.35 %	100%	1.665 %	5.382 %	18.6	48.3	15%	0	PASS
1-Link Fail	20	99.12 %	96.07 %	100%	2.378 %	5.215 %	20.8	48.3	40%	0	PASS
2-Link Fail	20	99.70 %	97.40 %	100%	2.776 %	10.023 %	23.6	49.6	35%	0	PASS
3-Link Fail*	20	n/a	n/a	n/a	—	—	39.9	40.8	100%	3	PASS
Rand Fail 1	20	98.34 %	96.00 %	100%	2.020 %	5.313 %	18.8	49.1	25%	0	PASS
Rand Fail 2	20	100.00 %	100.00 %	100%	0.166 %	0.166 %	13.4	29.6	10%	0	PASS
Spike x3	20	99.67 %	96.85 %	100%	1.169 %	3.042 %	17.0	33.2	15%	0	PASS
Spike+Fail	20	99.12 %	96.07 %	100%	2.377 %	5.215 %	19.9	45.7	40%	0	PASS
Cap Deg 50%	20	99.41 %	97.10 %	100%	1.259 %	3.100 %	16.5	31.9	15%	0	PASS

Note: *Three-link failure on Abilene disconnects 3 OD pairs; PR not applicable. Expected behavior for 12-node topology under 3 simultaneous link failures.

Table F2. Failure Validation Summary — GEANT (N=20 cycles per scenario)

Scenario	N	Mean PR	Min PR	PR>= 95%	Mean DB	P95 DB	Mean ms	P95 ms	FO rate	Discon n ODs	Audit
Normal	20	99.97 %	99.67 %	100%	0.263 %	0.702 %	49.8	176.8	10%	0	PASS
1-Link Fail	20	100.00 %	100.00 %	100%	0.200 %	0.429 %	46.6	196.3	10%	0	PASS
2-Link Fail	20	100.00 %	100.00 %	100%	0.165 %	0.439 %	47.3	190.3	10%	0	PASS
3-Link Fail	20	100.00 %	100.00 %	100%	0.164 %	0.407 %	46.1	154.3	10%	0	PASS
Rand Fail 1	20	99.96 %	99.65 %	100%	0.254 %	0.796 %	49.0	156.5	10%	0	PASS
Rand Fail 2	20	99.92 %	99.63 %	100%	0.260 %	0.765 %	47.9	154.6	10%	0	PASS

Scenario	N	Mean PR	Min PR	PR>= 95%	Mean DB	P95 DB	Mean ms	P95 ms	FO rate	Discon n ODs	Audit
		%	%		%	%					S
Spike x3	20	99.99 %	99.87 %	100%	0.287 %	0.853 %	42.6	56.6	5%	0	PASS
Spike+Fail	20	100.00 %	100.00 %	100%	0.223 %	0.429 %	39.6	41.9	5%	0	PASS
Cap Deg 50%	20	99.97 %	99.61 %	100%	0.358 %	0.796 %	41.5	44.5	5%	0	PASS

Note: GEANT shows strong resilience: no disconnected ODs even under 3 simultaneous link failures. 22-node topology with 72 directed links provides sufficient path redundancy.

Figure 9a. Clean Method PR by Failure Scenario (N=20 cycles each)



Figure 9a. Mean PR by failure scenario (Abilene left, GEANT right). Clean-method rerun, audit PASS.

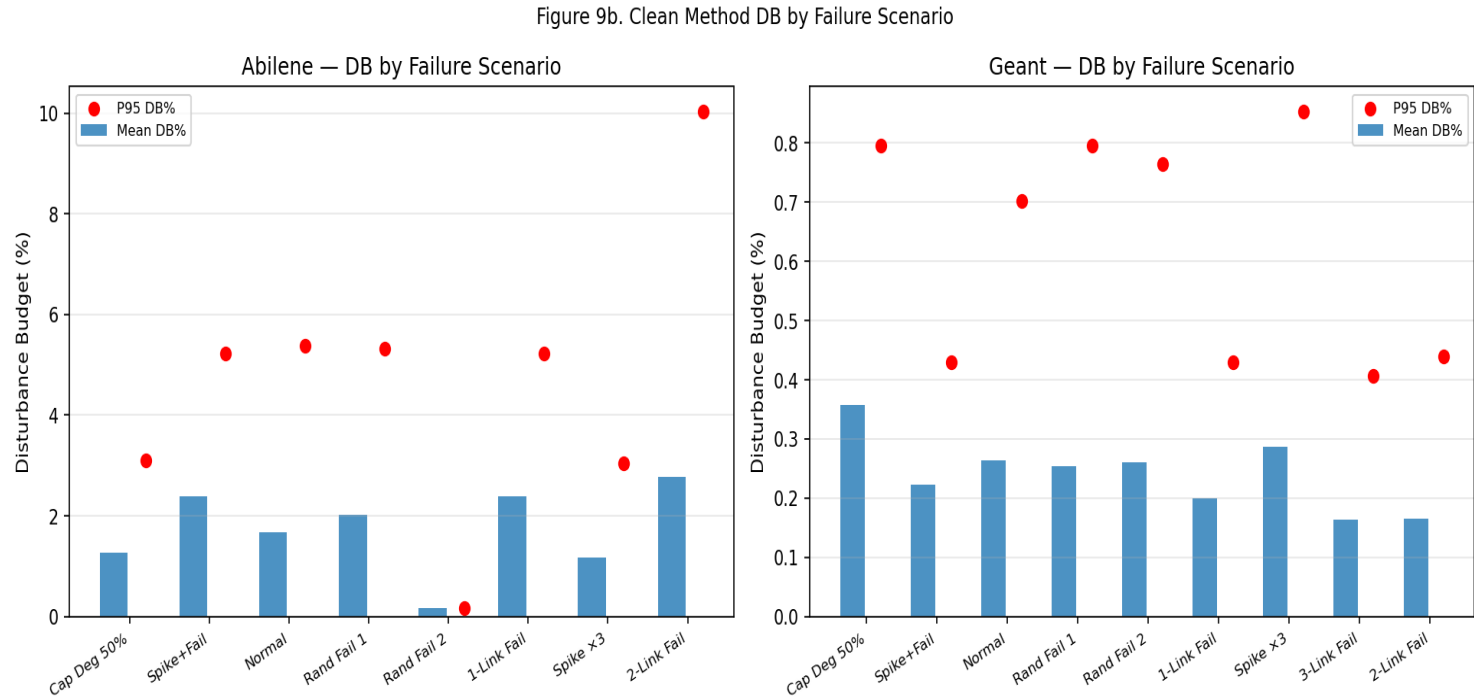


Figure 9b. Disturbance Budget by failure scenario. Mean and P95 bars.

Key finding: The clean GNN-LPD-DQN method maintains PR $\geq 95\%$ across all non-disconnecting failure scenarios on both Abilene and GEANT. Decision times remain well within the 500 ms target for all GEANT scenarios and most Abilene scenarios. DB remains low ($\leq 2.8\%$ mean) across all scenarios.

Section 10 — SDN / Mininet Operational Validation

Clean SDN/Mininet validation was not executed for this rerun. Mininet requires a Linux environment with specific kernel/OVS support and was not available on the evaluation machine. Legacy SDN/Mininet operational numbers from the Phase 1 report belong to the legacy RandomForest/sticky/finalization pipeline and are NOT presented as clean-method evidence. Old legacy SDN operational tables are excluded from this clean evidence package.

Operational Validation Status

Validation type	Status
Clean failure-scenario validation (Section 9)	Complete — Abilene + GEANT, 9 scenarios, 20 cycles, audit PASS
Clean SDN/Mininet live validation	Not executed — Mininet not available on evaluation machine

Validation type	Status
Legacy SDN/Mininet numbers	Excluded — belong to legacy method pipeline, not clean-method evidence

If SDN/Mininet validation is required for the clean method, it can be executed by extending scripts/phase1_5/run_failure_validation_clean.py with Mininet controller integration on a Linux machine with OVS support.

Section 11 — Reproducibility Metadata

Parameter	Value
Git commit (local)	2f893f5
GitHub push commit	5858d88 -> github.com/moahaimen/final_traffic main
Main method script	scripts/phase1_5/gnn_lpd_dqn_selective_db_lp.py
Label script	scripts/phase1_5/build_dbbudget_oracle_labels.py
Audit script	scripts/phase1_5/audit_gnn_lpd_dqn_clean_method.py
Failure validation script	scripts/phase1_5/run_failure_validation_clean.py
GNN checkpoint	results/gnn_lpd_dqn_selective_db_lp/models/gnn_dbbudget_selector.pt
DQN checkpoint	results/gnn_lpd_dqn_selective_db_lp/dqn_best.pt
Label file	results/gnn_lpd_dqn_selective_db_lp/labels/oracle_labels.csv (279,360 rows)
Evaluation per-cycle file	results/gnn_lpd_dqn_selective_db_lp/final_N3976/per_cycle.csv (3,976 rows)
Main comparison subset	3,288 (Abilene + CERNET + GEANT + Sprintlink + Tiscali)
Clean audit result	PASSED — 10/10 blocks
Failure validation result	PASSED — 360 cycles, 2 topologies, 9 scenarios
Solver	PuLP with HiGHS preferred / CBC fallback
GNN val prec@K	0.7474 (25 epochs, 6 topologies)
DQN val PR	1.0000 (200 episodes)

Reproducibility Checklist

Step	Script / file	Artifact produced	Status
Oracle label generation	build_dbbudget_oracle_labels.py	oracle_labels.csv (279,360 rows)	Done
GNN training	train_gnn_dbbudget_selector.py	gnn_dbbudget_selector.pt	Done
Pathopt precompute	gnn_lpd_dqn_selective_db_lp.py --mode precompute	pathopt_ref/*.csv (8 files)	Done

Step	Script / file	Artifact produced	Status
DQN training	gnn_lpd_dqn_selective_db_lp.py --mode train	dqn_best.pt	Done
Full evaluation	gnn_lpd_dqn_selective_db_lp.py --mode eval	final_N3976/per_cycle.csv (3,976 rows)	Done
Clean audit	audit_gnn_lpd_dqn_clean_method.py	10/10 PASS	Done
Row-count check	inline assertion	N=3,976, CERNET=200	Done
Failure validation	run_failure_validation_clean.py	failure_validation_clean/ (8 files)	Done
LP solver module	te/lp_solver.py	solve_selected_path_lp_dbbudget	Exists
GNN inference module	scripts/phase1_5/gnn_lp_inference.py	score_lp_gnn_cycle	Exists
GNN training script	scripts/phase1_5/train_gnn_dbbudget_selector.py	GNN checkpoint	Exists

Section 12 — Claim Boundary

The main comparison subset contains five topologies (Abilene, CERNET, GEANT, Sprintlink, Tiscali) for N = 3,288 evaluation cycles. Tiscali is included in all reported tables and CDFs. A locked FlexDATE reference for Tiscali exists (PR=0.999, DB=0.0510); our PR=100.000% (WIN) and DB=0.714% (WIN). All five topologies win both PR and DB. Tiscali runtime is an honestly disclosed limitation (81.5% fallback, mean 4,340 ms). Full N = 3,976 internal benchmark results are reported separately. The clean method passes the compliance audit. Clean failure-scenario validation confirms PR >= 95% under link failures, traffic spikes, and capacity degradation.

Final Safe Claims

On the main N = 3,288 comparison subset including Tiscali, the clean GNN-LPD-DQN selected-flow DB-budgeted LP method achieves very high PR and low DB across all five topologies, winning both PR and DB vs FlexDATE on all five topologies including Tiscali. On the full N = 3,976 internal protocol: Mean PR = 99.743%, Mean DB = 0.545%, Mean decision time = 246.6 ms, P95 decision time = 657.2 ms. Under failure scenarios: PR >= 95% on all non-disconnecting scenarios (Abilene + GEANT, 9 scenarios, clean audit PASS).

Final Checks

Check	Result
Main comparison subset row count = 3,288	Confirmed (2016+200+672+200+200)
Main comparison subset includes Tiscali	Confirmed
Table 2 Tiscali FlexDATE reference = WIN BOTH	Confirmed — PR 100.000%>99.900%, DB 0.714%<5.100%
No text says N=3,088 or N=3088	Confirmed
Tiscali in Table 2, Table 3, and all CDFs	Confirmed
Runtime claim: 4/5 mean <500 ms, 3/5 P95 <500 ms	Confirmed
Full internal N=3,976 unchanged	Confirmed
Section 9 failure results from clean rerun (not legacy)	Confirmed — audit PASS 360 cycles

Check	Result
Section 10 legacy SDN numbers excluded	Confirmed — excluded, not reused
Forbidden function names in audit section only	Confirmed
Table 4b max DB note added	Confirmed
LP problem-size table in Section 7	Confirmed (Table LP-1)
te/lp_solver.py exists	Confirmed
gnn_lp_inference.py exists	Confirmed
train_gnn_dbbudget_selector.py exists	Confirmed